

laser in question. There is much more than the power supply circuit that is used to operate the laser.

In the following paragraphs we shall take a closer look at the areas of electronics encountered in the case of three most commonly used laser sources, namely, solid-state, gas and semiconductor lasers. Prominent military laser systems configured around these laser sources are also briefly discussed.

Solid-state laser electronics

When we discuss electronics package of a flash-lamp-pumped pulsed solid-state laser (Nd-YAG, Nd-Glass, etc), it is the power supply unit which is needed to charge the energy storage capacitor to store the requisite quantum of energy that is considered representative of the electronics package. It is simply because of the reason that it is not only the most complex of the electronics circuit modules used in the case of flash-lamp-pumped solid-state lasers; its electrical conversion efficiency plays a key role in deciding the wall-plug efficiency and the size/weight of the laser. Capacitor-charging power supplies intended for flash-lamp-pumped solid-state laser applications are commercially available both as benchtop models as well as modular units for OEM applications. Fig. 1 shows photograph of one such capacitor-charging power supply suitable for flash-lamp-pumped solid-state lasers.

These capacitor-charging power supplies are available for a range of input voltage (AC or DC), DC output voltage and charging rate specifications. DC output voltage from a few hundreds of volts to several kilovolts and charging rate in the range of tens of joules per second to thousands of joules per second are common. In addition, these power supplies offer many control and protection features relevant to flash-lamp-pumped laser power supplies. Some of these features include end-of-charge status indication, peak output voltage hold, output voltage monitor, overvoltage and overtemperature protection and so on.

We also talk about, though with less

enthusiasm, the other circuit modules such as the simmer power supply which is invariably used in high-repetition rate, flash-lamp-pumped solid-state lasers or the Q-switch driver used in the Q-switched lasers, or even the pulse forming network (PFN) which ensures a critically damped current pulse through the flash lamp when the energy storage capacitor is made to discharge through it.

Nd-YAG, frequency-shifted Nd-YAG and erbium glass operating at



Fig. 2: Handheld eye-safe laser rangefinder

1064nm, 1540nm and 1550nm, respectively, are the most commonly used laser sources for laser rangefinding, target designation and laser-guided munitions delivery applications. While Nd-YAG is almost invariably used for target designation and laser-guided munitions delivery, lasers operating at 1540/1550nm are preferred for rangefinding applications considering the eye safety of operating personnel.

Semiconductor diode lasers are also used for rangefinding applications both for commercial and military domains. Their use is mainly restricted to relatively shorter ranges with maximum measurable range generally not exceeding 5km. On the other hand, portable and handheld Nd-YAG laser-based rangefinders are available for maximum range measuring capability in excess of 25km. Fig. 2 shows the photograph of one such handheld laser rangefinder.

In addition to conventional target rangefinding, there are many related applications where the basic rangefinding concept is put to use in different

military laser systems. Some of the prominent ones include proximity sensors, obstacle avoidance sensors and bathymetry for sea-bed mapping. These devices also make use of either solid-state or semiconductor diode lasers.

Semiconductor diode laser electronics

Design of drive and control circuits needed to power semiconductor diode lasers should consider certain handling and protection issues if they were to have the prescribed life and reliability performance. It is more so for semiconductor diode lasers used in military applications. Diode lasers are particularly sensitive to electrostatic discharge, short-duration electric transients such as current spikes, injection current exceeding the prescribed limit and reverse voltage exceeding the breakdown limit.

In order to protect the diode lasers from above failure modes, the driver circuit should be carefully designed and should have all the features recommended by the diode-laser manufacturer. The driver should be a constant current source with in-built features such as soft start, protection against transients, interlock control for the connection cable to the laser and safe adjustable limit for injection current.

In case the laser is to be operated in pulsed mode, the injection current should be pulsed between two values above the lasing threshold rather than between cut-off and lasing mode. A laser diode when used in a laser printer or a laser pointer, or even a compact disk player, may need a conventional constant-current source without too stringent a requirement on the current stabilisation to do the job.

Drive current and diode temperature stabilisation to a high degree become extremely important when the intended application demands a stable output wavelength. One such application area is in laser-based Raman sensor used for detection and identification of chemical warfare agents and explosive materials. The concepts of drive current and diode temperature